

Chapter 2 Cardiovascular examination

Although technology has a high profile in **cardiology**, clinical examination remains a central tool, especially for the generalist.

General inspection

Many clues to the cardiac condition can be detected with a simple visual inspection. In the acutely unwell patient, cyanosis, pallor, and sweatiness can all be signs of impending danger – does the patient "look" ill? In nonacute patients, cachexia is perhaps the most important feature to note on general inspection since it is an important prognostic sign in heart failure. Palpation is essential to confirm that girth is excess fluid (pitting edema). Certain physical appearances should always prompt an awareness of cardiac abnormalities (see Table 1). Facial signs for which there is evidence of an association with cardiac conditions are shown in Table 2. Finally, it is important to document the condition of a potential cardiac patient's teeth.

Taking the pulse

Taking the pulse is one of the simplest, oldest, and yet most informative of all clinical tests. As you pick up the patient's hand, you should check for clubbing and any peripheral signs of endocarditis (see Table 3). Note the rate and document the rhythm of the pulse. The character and volume of the pulse can also be useful signs and traditionally it is believed that these are easier to detect in larger arteries such as the brachial and the carotid (see Table 4).

Checking both radials simultaneously is important in all cases of chest pain as a gross screening test for aortic dissection. Adding radiofemoral delay (or radiofemoral difference in volume) may alert you to coarctation as a rare cause of hypertension.

Peripheral pulses should also be documented, as peripheral vascular disease is an important predictor of coronary artery disease:

- femoral – feel at the midinguinal point (midway between the symphysis pubis and the anterior superior iliac spine, just inferior to the inguinal ligament)
- popliteal – feel deep in the center of the popliteal fossa with the patient lying on their back with their knees bent
- posterior tibial – feel behind the medial malleolus
- dorsalis pedis – feel over the second metatarsal bone just lateral to the extensor hallucis tendon

Blood pressure

This is described in [Chapter 6](#), Hypertension.

Jugular venous pressure

Of all the elements of clinical examination, the jugular venous pressure (JVP) is the most mysterious. It is highly esoteric, and whilst some people wax lyrical about the steepness of the "y" descent, others will feel grateful to be convinced they see it at all. Two things are very clear: (1) the JVP is a very useful clinical marker in many situations, and (2) the exact height of the JVP is a poor guide to central venous pressure. Taken together, this suggests that noting whether the JVP is "up" or "down" is good practice in every cardiac patient. In particular, it can be very useful in diagnosing right-sided heart failure and in differentiating a cardiovascular cause of acute shortness of breath (right ventricular failure, pulmonary embolism) from an intrinsic pulmonary cause (asthma, chronic obstructive pulmonary disease). For the general physician, the waveform of the JVP (see Figure 1) is, for most purposes, only of academic significance.

The JVP should be assessed with the patient reclined at a 45° angle (see Figure 2). Accepted practice is that only the internal jugular vein should be used, as only this vessel joins the superior vena cava at a 180° angle. The JVP is defined as the height of the waveform in centimeters above the sternal angle (<4 cm is normal). Abnormalities of the JVP are described in Table 5.

Palpation

Before auscultation, inspection of the precordium can be a useful indicator of previous surgery – eg, midline sternotomy suggests previous bypass, lateral thoracotomy suggests previous mitral valve or minimally invasive bypass surgery (left internal mammary artery to left anterior descending coronary artery). Locate the apex beat – the furthest point laterally and inferiorly where you can clearly feel the apex (usually the fifth intercostal space in the midclavicular line). There are many different descriptions for abnormal apex beats. One scheme distinguishes heaving (high afterload, eg, aortic stenosis) from thrusting (high preload, eg, aortic regurgitation). The apex may also be "tapping", but this reflects a loud first heart sound. In addition, you should place your left hand over the sternum and feel for any significant ventricular heave (right ventricular hypertrophy) or thrill (tight aortic stenosis, ventricular septal defect).

Auscultation

Held by many as the key to physical examination, the importance of auscultation remains, but is diminished in an age of increasingly portable echocardiography.

Listen over the aortic (second right intercostal space) and pulmonary (second left intercostal space) areas and at the left lower sternal edge with the diaphragm of your stethoscope (better for higher pitches), then use the bell for the apex (better for lower pitches). If in doubt, use both. Press lightly with the bell. If you hear an abnormality over the aortic or pulmonary areas, you should listen over the carotids. If you hear an abnormality at the apex, listen in the axilla. Listen systematically. Start with the heart sounds – ignore everything else.

Heart sound variations

When listening to heart sounds, note their volume (normal, diminished, loud) and whether physiological splitting is present (see Figure 3).

Physiological splitting of the second heart sound is when the sound of aortic valve closure (A2) occurs earlier than that of pulmonary valve closure (P2). It occurs in inspiration and is more common in the young. It is caused by increased venous return and negative intrathoracic pressure. This delays right ventricular emptying and pulmonary valve closure, at the same time that pooling of blood in the pulmonary capillary bed hastens left ventricular emptying and aortic valve closure. Reverse splitting of the second heart sound can occur in conditions where aortic valve closure is delayed, such as left bundle branch block or paced right ventricle, or where pulmonary valve closure occurs early, such as in the B form of Wolff–Parkinson–White syndrome. Wide fixed splitting of the second heart sound occurs in atrial septal defect.

A third heart sound may be heard soon after the second heart sound. It is thought to be due to rapid, high-volume filling of the left ventricle. As such, it is found in pathological (left ventricular failure) as well as physiological (athletic heart, pregnancy) states.

A fourth heart sound may be heard just before the first sound. This is caused by atrial contraction filling a stiff left ventricle, eg, hypertensive heart or diastolic heart failure.

Murmurs

When you have considered these heart sound variations, move on to consider the gaps between the heart sounds. If you hear a murmur, first establish whether it occurs in systole or diastole (time against the carotid pulse if necessary). Then determine its length and, if short, its exact position (early, mid, or late; systole or diastole) (see [Figure 3](#)).

Added sounds

An opening snap occurring after the second heart sound represents a diseased mitral valve opening to a stenotic position. An ejection click soon after the first heart sound occurs in aortic stenosis and pulmonary stenosis. A mid systolic click is heard in mitral valve prolapse.

After listening to the heart

Listen to the base of the lungs for the fine inspiratory crackles of pulmonary edema. If you suspect right-sided cardiac pathology, palpate the liver, which will be enlarged, congested, and possibly pulsatile in cases of right ventricular failure or tricuspid valve disease. Also, check the patient's ankles for swelling.

[Table 6](#) outlines common associations in cardiovascular clinical examination.

Summary

A careful clinical examination can reveal much about the condition of your patient's heart. In addition, noting the findings of a full examination will greatly facilitate specialist referral. In an age of high technology, skilled clinical examination has yet to be surpassed in terms of convenience, safety, and value for money.

Further reading

1. Bickley LS, Hoekelman RA, editors. *Bates' Pocket Guide to Physical Examination and History Taking*, 3rd edn. Lippincott Williams & Wilkins, 2000.
2. Gleadle J. *History and Examination at a Glance*. Blackwell Science, 2003.
3. Perloff JK. *Physical Examination of the Heart and Circulation*, 3rd edn. WB Saunders, 2000.
4. Turner RC, Blackwood RA. *Lecture Notes on Clinical Skills*, 3rd edn. Blackwell Science, 1997.

Figures

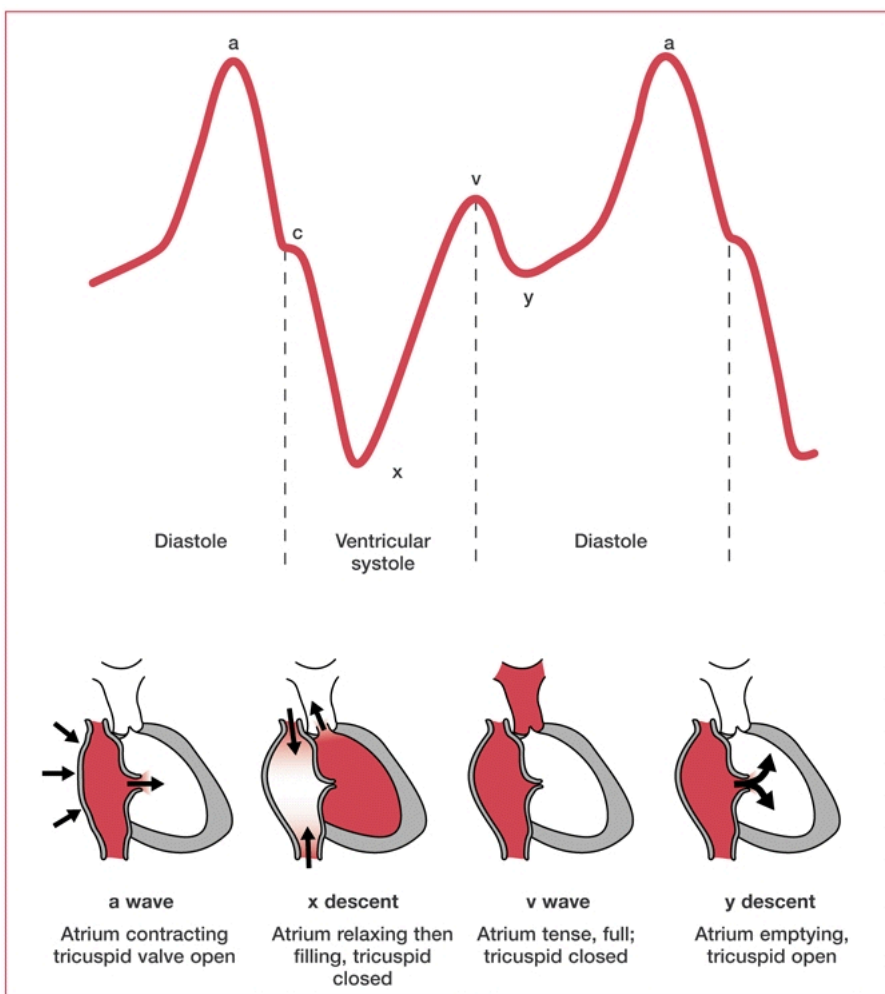


Figure 1

Waveforms of the jugular venous pressure (including a brief explanation for each wave). The "c" wave represents right ventricular contraction "pushing" the tricuspid valve back into the right atrium. Reproduced with permission from Oxford University Press (Longmore JM et al. *The Oxford Handbook of Clinical Medicine*, 5th Edn, p. 79).

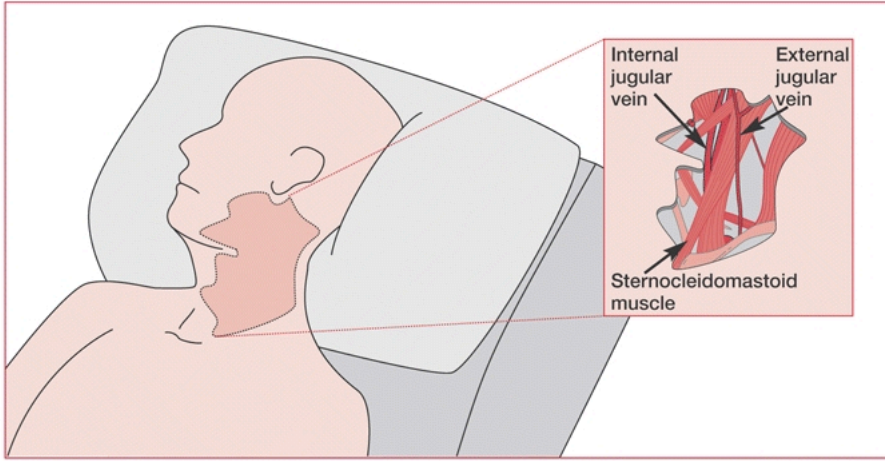


Figure 2

The jugular veins. The patient is lying at a 45° angle, thus revealing the surface markings of the neck.

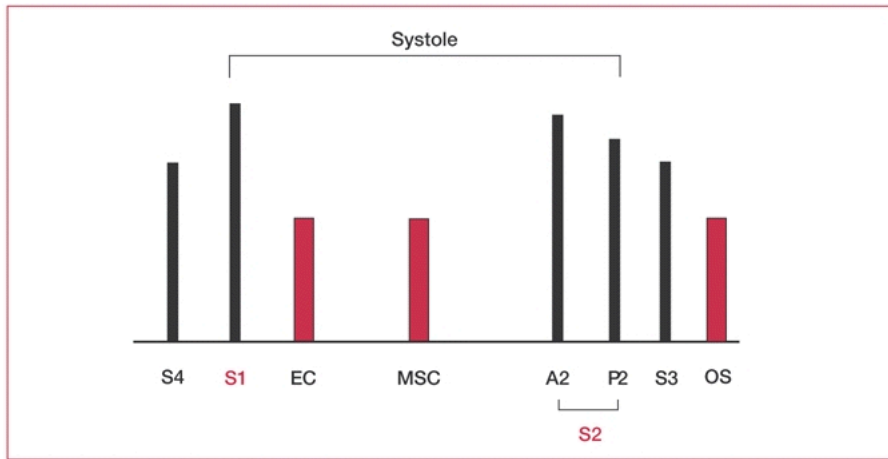


Figure 3

Relative positions of heart sounds and added sounds in auscultation. Sounds in red are high pitched. A2: aortic component of second heart sound; EC: ejection click; MSC: mid systolic click; OS: opening snap; P2: pulmonary component of second heart sound; S1–S4: heart sounds 1–4.

Tables

Table 1 Cardiac manifestations of genetic disorders

Genetic disorder	Associated cardiac manifestation
Marfan's syndrome	Aortic regurgitation (aortic dissection)
Down's syndrome	ASD, VSD
Turner's syndrome	Coarctation of the aorta
Spondyloarthritides, eg, ankylosing spondylitis	Aortic regurgitation

ASD: atrial septal defect; VSD: ventricular septal defect.

Table 2 Facial signs associated with cardiac conditions

Facial sign	Description	Possible cardiac association
Malar flush	Redness around the cheeks	Mitral stenosis
Xanthomata	Yellowish deposits of lipid around the eyes, palms, or tendons	Hyperlipidemia
Corneal arcus	A ring around the cornea	Age, hyperlipidemia
Proptosis	Forward projection or displacement of the eyeball; occurs in patients with Graves' disease	Atrial fibrillation

Table 3 Peripheral signs associated with infective endocarditis

Peripheral sign	Description	Cardiac association
Clubbing	Broadening or thickening of the tips of the fingers (and toes) with increased lengthwise curvature of the nail and a decrease in the angle normally seen between the cuticle and the fingernail	Infective endocarditis, cyanotic congenital heart disease
Splinter hemorrhages	Streak hemorrhages in nailbeds	Infective endocarditis
Janeway lesions	Macules on the back of the hand	Infective endocarditis
Osler's nodes	Tender nodules in fingertips	Infective endocarditis

Table 4 Abnormal pulses

Type of pulse	Pulse characteristics	Most likely cause
Regularly irregular	–	2nd-degree heart block, ventricular bigeminy
Irregularly irregular	–	Atrial fibrillation, frequent ventricular ectopics
Slow rising	Low gradient upstroke	Aortic stenosis
Waterhammer, collapsing	Steep up and down stroke (lift arm so that wrist is above heart height)	Aortic regurgitation, patent ductus arteriosus
Bisferiens	A double-peaked pulse – the second peak can be smaller, larger, or the same size as the first	Aortic regurgitation, hypertrophic cardiomyopathy
Pulsus paradoxus	An exaggerated fall in pulse volume on inspiration (>10 mm Hg on sphygmomanometry)	Cardiac tamponade, acute asthma
Bounding	Large volume	Anemia, hepatic failure, type 2 respiratory failure (high CO ₂)
Pulsus alternans	Alternating large and small volume pulses	Bigeminy

Table 5 Abnormalities of the jugular venous pressure (JVP)

JVP abnormalities	Probable cause
Large "a" wave	Tricuspid stenosis, pulmonary hypertension, pulmonary stenosis
Cannon wave	Atrial fibrillation, complete heart block, VVI pacing, ventricular tachycardia (a cannon wave occurs when the right atrium contracts against a closed tricuspid valve)
Steep "x", "y" descent	Constrictive pericarditis, cardiac tamponade
Large "v" wave, "cv" wave	Tricuspid regurgitation
Kussmaul's sign	Rise of JVP on inspiration, constrictive pericarditis, cardiac tamponade

Table 6 Common associations in cardiovascular clinical examination

	AS	Aortic sclerosis/minimal AS	Aortic regurgitation	Mitral stenosis	Mitral regurgitation	Tricuspid regurgitation	Pulmonary regurgitation	Tricuspid stenosis	Pulmonary stenosis	ASD	VSD	PDA
Pulse	Low volume, slow rising	Normal	↑ volume, collapsing	Low volume, normal/AF	Normal/AF	Normal/AF	Normal	Usually low volume, AF	Low volume	Normal/AF	Normal	Reduced
Pulse pressure	↓	—	↑	—	—	—	—	—	—	—	—	↑
JVP	—	—	—	—	—	↑, prominent systolic wave ("cv" wave)	—	Prominent "a" wave (if sinus rhythm)	Large "a" wave	—	—	—
Apex	Heaving, not displaced	Just palpable, not displaced	Thrusting, displaced	Tapping, not displaced	Thrusting, displaced	—	—	—	—	Just palpable, not displaced	May be displaced	Third point
First sound	—	—	—	Loud	Soft	—	—	—	—	—	—	—
Second sound	Soft A2	A2 not soft	—	—	—	—	—	—	Soft P2	Wide fixed splitting of S2	P2 may be loud	—
Added sounds	Ejection click can occur with bicuspid valve	—	—	No third sound opening snap	Third sound	—	—	—	—	—	—	—
Murmur	Loud, harsh, mid-systolic ejection	Ejection systolic, neither harsh nor loud	Blowing, high-pitched early diastolic	Low, rumbling, mid-diastolic	Pansystolic	Pansystolic	Early diastolic	Rumbling mid-diastolic	Harsh mid-systolic ejection	Ejection systolic (↑ flow across pulmonary valve) ± harsh, explosive, brief early diastolic (PR) and ejection click	Pansystolic ± early diastolic (PR)	Continuous murmur with systolic accentuation
Heard best	Second right ICS	Second right ICS	LLSE (patient forward in expiration)	Apex with patient turned to left	Apex	LLSE in inspiration	Second left ICS	LLSE (louder on inspiration)	Second left ICS (louder on inspiration)	—	LLSE ± second left ICS	5th left intercostal space
Radiates	Into carotids	Faintly to carotids	—	—	To axilla	—	—	—	—	—	Apex	Posteriorly

AF: atrial fibrillation; AS: aortic stenosis; ASD: atrial septal defect; ICS: intercostal space; JVP: jugular venous pressure; LLSE: left lower sternal edge; PDA: patent ductus arteriosus; PR: pulmonary regurgitation; VSD: ventricular septal defect. See Chapter 9, Valve disease, for more on columns 1–9 and Chapter 14, Adult congenital heart disease, for columns 10–13.